CS 120 — INTRODUCTION TO COMPUTER SYSTEMS

Module 1

Digital Citizenship:

At Home, School & Work

Computer Science Department · School of ICT · Copperbelt University

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LECTURE OVERVIEW — Module 1: Digital Citizenship

Five sections across three contexts of digital life:

A — **At Home**

Digital Citizen · DIKW · Ribble's 9 Elements · Digital Divide · Technology & Society · Identity Theft · Online Safety · Misinformation · Netiquette

B — **At School**

Academic Integrity · Plagiarism (7 types) · AI Tools Policy · Acceptable Use Policy

C — **At Work**

Technology in Professional World · Digital Presence · Workplace ICT Policies · Distracted Driving

D — **Digital Footprint & Privacy**

Active vs. Passive Footprint · Privacy Management · Zambian Law · Green Computing · Assistive Technologies

E — **Zambian Legal Framework**

ICT Act 2009 · Cyber Security & Cyber Crimes Act 2021 · Data Protection Act 2021

What is a Digital Citizen?

A digital citizen is a person who uses information and communication technology responsibly, safely, and ethically — while exercising informed awareness of their rights and responsibilities in digital spaces.

Digital LITERACY	Digital CITIZENSHIP
▸ Knowing HOW to use technology	▸ Knowing WHEN and WHETHER to use technology
▸ Technical skills: copy-paste, search, browse	▸ Ethical, legal, and responsible use
▸ Understanding what devices and software do	▸ Protecting yourself and others online
▸ Ability to operate digital tools effectively	▸ Contributing positively to digital society

Key Distinction: Digitally literate people know HOW to copy and paste information. Digital citizens know WHEN it is appropriate to do so, how to properly credit the source, and the ramifications of violating copyright. (Campbell, 2023)

Ribble's Nine Elements of Digital Citizenship

Mike Ribble (2015) identified nine distinct elements that together constitute responsible digital citizenship. ALL NINE are examinable.

1
Digital Access
Equitable participation for ALL in digital society

2
Digital Commerce
Safe, legal online buying and selling (mobile money)

3
Digital Communication
Responsible electronic exchange of information

4
Digital Literacy
Effectively using and critically evaluating technology

5
Digital Etiquette
Norms of acceptable conduct in digital spaces

6
Digital Law
Legal rights and responsibilities of technology users

7
Rights & Responsibilities
Freedoms and obligations balanced in digital spaces

8
Digital Health
Physical and psychological wellbeing in digital life

9
Digital Security
Protecting devices, data, and identity from harm

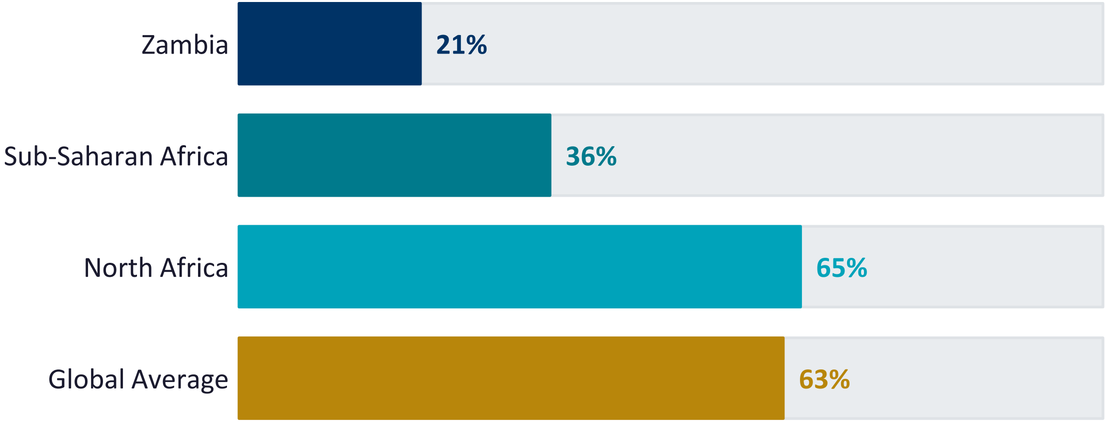
The Digital Divide — Zambian Context

The digital divide is the gap between those who have access to modern ICT and the internet and those who do not. It is multidimensional — not just about physical access.

21%
of Zambians use the internet

48%
global internet penetration avg

80%+
Zambia internet access via mobile



Contributing Factors

- Geography — rural vs. urban connectivity gap
- Cost — data bundles expensive relative to income
- Gender — women less likely to own smartphones
- Infrastructure — load-shedding disrupts access
- Skills — limited digital training outside cities
- Language — limited Zambian language digital content

Source: ITU ICT Development Index 2023 · ZICTA Annual Report 2023

How Technology Shapes Daily Life — Convergence

Convergence: The increasing integration of technological capabilities into a growing number of previously unrelated devices. You access the same email, social networking, banking, and apps on your laptop, tablet, and smartphone.

🏥 Healthcare

- mHealth apps monitor chronic conditions remotely
- Telemedicine connects rural patients to doctors via video call
- Wearable blood pressure & glucose monitors send data to clinics
- 3D printing creates prosthetics and surgical models

🏭 Manufacturing & Logistics

- CAM (Computer-Aided Manufacturing) uses robots for precision tasks
- Machine-to-machine communications monitor assembly lines
- GPS tracks delivery vehicles; handheld scanners log packages
- Sensors detect machine failures before they occur (predictive maintenance)

🎓 Education

- Learning Management Systems (Moodle, Teams) track student progress
- Interactive whiteboards replace traditional blackboards
- Online collaborative tools enable group projects across campuses
- Video lectures make education accessible offline

🛒 Retail & Finance

- IoT sensors track inventory in real-time
- Mobile money (MTN MoMo, Airtel) replaces cash transactions
- E-commerce enables buying without visiting physical stores
- Data analytics predict customer behaviour and demand

MODULE 1 · SECTION A

Digital Citizenship at Home

Screen Time · Privacy · Online Safety · Misinformation · Netiquette

Screen Time & Digital Wellness

6.5 hrs

Average daily screen time (DataReportal, 2024)

✅ Healthy Digital Habits

- Set daily screen time limits (Android: Digital Wellbeing | iOS: Screen Time)
- Keep devices out of bedrooms — charge phones in the lounge
- Take a 5-min movement break every 45–60 minutes
- Apply 20-20-20 rule consistently during computer work
- Use Focus/Do Not Disturb during study and family time

Physical Health Risks

👁 Computer Vision Syndrome (CVS): Eye strain, dry eyes, headaches → Apply 20-20-20 Rule: every 20 mins, look 20 feet away for 20 seconds

⌨ Repetitive Strain Injury (RSI): Tendon inflammation from keyboard and mouse use → ergonomic setup, regular breaks

🔙 Tech Neck: Poor posture from looking down at phones → screen at eye level; take standing breaks

😴 Sleep disruption: Blue light suppresses melatonin → avoid screens 1 hr before bed; use Night Mode

🏃 Sedentary risk: Prolonged sitting linked to cardiovascular disease → take a movement break every 45–60 mins

Psychological Health Risks

📱 Social media anxiety: Heavy social media use linked to depression, FOMO, and comparison culture

🎮 Digital addiction: Platforms designed to be compulsive — notice when use is habitual vs. intentional

🛡 Cyberbullying: Online harassment causing real psychological harm — criminal under Zambia Cyber Crimes Act 2021

Online Safety for the Family

Children & Teenagers

- Apply 'stranger danger' online — never share name, school, address, or photos with unknown contacts
- Use parental controls: Google Family Link (Android) or Apple Screen Time (iOS)
- Age-appropriate platforms: Facebook, Instagram, TikTok all require age 13+
- Keep devices in common areas — bedrooms increase risk of unsupervised use
- Know how to report cyberbullying — to a trusted adult, school, and ZICTA
- Online predators exist in Zambia — ZICTA handles reports: cybercrime@zicta.zm

Parents & Older Adults

- Beware the 'grandparent scam': fake emergency calls demanding urgent mobile money
- NEVER transfer mobile money to someone you cannot independently verify
- WhatsApp forwards: treat health claims, government announcements, and competitions with SIFT scrutiny
- Enable Two-Step Verification on WhatsApp and banking apps to prevent account hijacking
- Legitimate organisations never send urgent payment requests via WhatsApp
- If a deal seems too good to be true on social media — it is a scam

🔐 **ZICTA Cybercrime Reporting: +260 211 221 220 | cybercrime@zicta.zm | www.zicta.zm**

Identity Theft & Protecting Your Privacy

Privacy: The state of being free from public attention to the degree that you determine. Your right to be left alone online.
Identity Theft: Using someone's personal information (NRC number, bank account, passwords) to commit financial fraud.

Table 1-1: How Personal Information is Stolen (Campbell, 2023)

Phishing:	Fraudulent emails or websites trick you into entering passwords, PINs, or card numbers. Very common via fake MTN/Airtel messages in Zambia.
Pretexting:	Attacker pretends to be from a legitimate organisation (bank, ZICTA, ZRA) and requests personal information.
Dumpster Diving:	Personal info recovered from discarded bank statements, receipts, or documents. Shred sensitive paperwork.
Data Mining:	Attackers guess security questions using info you posted on social media (pet's name, hometown, mother's maiden name).
Change of Address:	Fraudster redirects your mail to intercept bank statements, OTP codes, and account information.
SIM Swapping:	Attacker convinces your mobile provider to transfer your number to a new SIM — then intercepts your SMS verification codes.

✓ PROTECT YOURSELF: Never share PINs/passwords via phone or email · Shred sensitive documents · Monitor mobile money statements weekly · Enable 2FA on all accounts · Use unique passwords per service

Misinformation, Disinformation & the SIFT Framework

False news spreads 6× faster than true news on social media. (Vosoughi, Roy & Aral, 2018)

Misinformation

False or inaccurate information shared WITHOUT intent to deceive. The sharer genuinely (but incorrectly) believes it is true.

📍 *e.g. Sharing a WhatsApp forward about a herbal COVID cure because you believe it helps people.*

Disinformation

False information DELIBERATELY created and shared with intent to deceive, manipulate, or cause harm.

📍 *e.g. A fabricated news story claiming the government has frozen mobile money accounts to cause panic withdrawals.*

Malinformation

Information that is FACTUALLY TRUE but shared with intent to harm — e.g. leaking private photos or confidential records to damage someone.

📍 *e.g. Sharing someone's private medical records or intimate images to ruin their reputation.*

The SIFT Framework for Verification — Apply BEFORE sharing (Caulfield, 2019)

S — STOP

Pause before sharing or reacting. Ask: am I feeling unusually emotional? That is a manipulation signal.

I — INVESTIGATE the source

Who made this? What is their track record? Is the URL a fake lookalike domain (e.g. 'ZambiaNews.net' vs 'lusakatimes.com')?

F — FIND better coverage

Do multiple independent, credible sources report the same story? If only one source carries it, be very cautious.

T — TRACE claims & images

Use reverse image search (Google Images) to find the original context of a photo. Check publication dates.

Netiquette — The Rules of Internet Etiquette

Netiquette: The code of acceptable behaviours that govern how individuals should conduct themselves while using the internet — covering email, social media, messaging, online discussions, and file sharing. (Campbell, 2023)

Email (Professional)

DOs	DON'Ts
✓ Use a descriptive subject line	✗ Use emojis, abbreviations (lol, btw)
✓ Address formally (Dear Dr. Mwansa,)	✗ CC unnecessary people
✓ Be concise — state purpose in first sentence	✗ Use ALL CAPS (it reads as shouting)
✓ Proofread before sending	✗ Send angry emails — draft, wait, review
✓ Reply within 24–48 hours	✗ Use personal Gmail for formal submissions
✓ Use professional email signature	

Social Media & Messaging

DOs	DON'Ts
✓ Be respectful even when disagreeing	✗ Post anything you would not say face-to-face
✓ Verify before sharing (apply SIFT)	✗ Share others' private conversations or photos
✓ Give credit when sharing others' content	✗ Engage in pile-ons or public shaming
✓ Use trigger warnings for sensitive content	✗ Use fake accounts to attack or mislead
✓ Respect others' privacy settings	✗ Tag people in embarrassing content

MODULE 1 · SECTION B

Digital Citizenship at School

Academic Integrity · Plagiarism · AI Tools · AUP

Academic Integrity — The Six Pillars (ICAI, 2021)

Academic integrity is the foundation of your degree's value. Without it, your CBU qualification means nothing — to you or to employers.

HONESTY

Represent your own work accurately. Give credit where credit is due. Never misrepresent data or sources.

TRUST

Build a learning community where everyone can rely on each other to act with integrity consistently.

FAIRNESS

Clear, equitable standards apply to all. Everyone competes under the same rules.

RESPECT

Honour others' ideas and intellectual property — classmates, lecturers, authors, and researchers.

RESPONSIBILITY

Own your choices. Do not facilitate others' dishonesty. Report violations when you observe them.

COURAGE

Do the right thing even when it is uncomfortable. Cite unpopular sources. Challenge unfair practices.

Violations include: plagiarism (any form) · contract cheating · exam misconduct · fabricating data · impersonation · falsifying academic records · using AI without authorisation. **Consequences:** fail the assessment → fail the module → academic suspension → expulsion → possible criminal liability.

Plagiarism — Types, Severity & How to Avoid It

Plagiarism can occur with or without intent to deceive — ignorance is not a defence.

Type	Description	Severity
Direct Copy:	Verbatim reproduction without quotation marks or citation	CRITICAL
Paraphrase Plagiarism:	Rewording someone's ideas without crediting the original source	HIGH
Mosaic / Patchwork:	Blending copied phrases from multiple sources without attribution	HIGH
Contract Cheating:	Paying/persuading someone else to produce your work	CRITICAL
Self-Plagiarism:	Resubmitting your own previous work without disclosure	MEDIUM
Incorrect Citation:	Misrepresenting what a source actually says	MEDIUM
Accidental:	Omitting a citation due to poor note-taking during research	LOW–MED

✓ HOW TO AVOID: Always cite when paraphrasing · Use quotation marks for verbatim text · Record sources AS you research · Follow APA 7th Edition · Use Turnitin before submission · When in doubt, CITE.

Artificial Intelligence Tools in Academic Work

CBU AI Policy for Academic Submissions is under development. Until module-specific guidance is issued, use this framework:

✅ GENERALLY PERMITTED

No special disclosure required

- Grammar and spelling check (Grammarly, Word Editor)
- Brainstorming and exploring topics before writing
- Getting explanations of complex concepts you then verify
- Translating content to help understand a foreign-language source
- Generating boilerplate code you FULLY UNDERSTAND and modify
- Summarising sources (which you then read and verify yourself)

⚠️ PERMITTED WITH DISCLOSURE

Declare AI use in your submission; describe specifically how and what you changed

- AI-assisted drafting that you then substantially revise and verify
- Using AI to suggest structure or outline — that you then develop independently
- AI-generated code as a starting point — that you understand, test, and explain

🚫 NOT PERMITTED — MISCONDUCT

Treated as academic misconduct — same consequences as plagiarism

- Submitting AI-generated text as your own writing
- Using AI to answer exam or quiz questions
- Asking AI to fabricate or invent references
- Submitting AI-generated code you cannot explain

⭐ **GOLDEN RULE: If you cannot explain every part of your submitted work without AI assistance, you have likely violated academic integrity. AI is a thinking AID, not a thinking REPLACEMENT.**

A typical University ICT Acceptable Use Policy (AUP)

An AUP is a document listing guidelines and consequences for using institutional ICT resources, including networks, email, and storage. AUP must be read and signed before accessing any university ICT resources.

✅ You MAY	🚫 You MAY NOT
✓ Use university ICT for academic work, research, and educational internet use	✗ Download, store, or share pirated software, music, or films
✓ Access the LMS (Moodle) and submit assignments online	✗ Access pornographic, violent, or illegal content on CBU networks
✓ Use student email for official communication	✗ Attempt to bypass, test, or exploit security controls or firewalls
✓ Access online journals and databases from the library	✗ Use bandwidth for cryptocurrency mining or non-academic torrenting
✓ Print study materials in the computer labs	✗ Access or modify other users' files, emails, or accounts
✓ Collaborate with classmates on group projects via LMS	✗ Use CBU infrastructure for commercial activities without permission

Consequences: Written warning + ICT suspension (first offence) → Disciplinary Committee referral (repeat/serious offence) → Academic suspension/expulsion → Referral to police under the Cyber Security & Cyber Crimes Act 2021 (hacking, fraud).

MODULE 1 · SECTION C

Digital Citizenship at Work

Technology in Professional World · Digital Presence · ICT Policies · Distracted Driving

Technology in the Professional World

An intelligent workplace connects employees to networks, productivity tools, conferencing, and real-time data. Every major sector in Zambia is being transformed by technology.

Healthcare — mHealth & Telemedicine

- mHealth: Doctors use tablets/smartphones to access patient records stored on the cloud — reducing unnecessary clinic visits
- Telemedicine: Secure video consultations connect rural Zambian patients to specialists in Lusaka without travel
- Wearable monitors track blood pressure and glucose, sending alerts to healthcare providers automatically
- Zambia example: UTH and private clinics using WhatsApp-based telemedicine pilots during and after COVID-19

Manufacturing — CAM & Automation

- Computer-Aided Manufacturing (CAM) streamlines production and ships products faster with fewer errors
- Robots perform tasks too dangerous, detailed, or monotonous for humans — precision welding, chemical handling
- Machine-to-machine (M2M) communications monitor assembly lines without human intervention
- Zambia example: Copperbelt mine automation — automated drilling systems and underground robot-assisted operations

Transport & Logistics

- Handheld computers scan barcodes/QR codes on packages before loading for delivery
- AI routing software finds the most efficient delivery path, accounting for traffic and road conditions
- GPS navigation guides drivers; telematics monitors driver hours and vehicle safety remotely
- Zambia example: ZAMPOST and private couriers using scan-track-deliver systems for parcel tracking

Your Professional Digital Presence

70%

of employers screen candidates' social media BEFORE hiring

BUILD YOUR DIGITAL BRAND NOW — Your online presence begins the moment you become a CBU student. Future employers will search your name. What will they find? Start shaping your professional digital identity today.

LinkedIn Profile

- Professional headshot (not a selfie or party photo)
- Clear headline: 'Computer Science Student at CBU | Interested in Software Development'
- List CBU education, ALL modules, and notable projects
- Connect with lecturers, classmates, and professionals in your field
- Engage thoughtfully — share industry articles, comment on posts

GitHub Portfolio

- Create an account and commit your first project this semester
- A populated GitHub profile shows consistent technical activity to recruiters
- Readme files: document your projects clearly — what they do, how to run them
- Open-source contributions: even small contributions count
- CBU projects become portfolio pieces — take them seriously

Email Etiquette: Descriptive subject line · Formal greeting (Dear Dr. Surname) · Concise purpose in first sentence · Proofread · Reply within 24–48hrs · Professional signature (Name, Programme, Year, CBU) · NO emoji or abbreviations

Workplace ICT Policies — BYOD, Confidentiality & Collaboration

Most organisations with significant ICT infrastructure operate under a formal ICT policy. As a new employee you will be required to read and sign this policy before accessing any systems.

BYOD — Bring Your Own Device

- BYOD policies allow employees to use personal devices (laptops, phones) for work — reducing costs but increasing security risks
- RISK: Personal and corporate data on the same device; loss exposes company data; personal apps may leak corporate data
- SAFE BYOD: Screen lock + strong PIN · Full-disk encryption · MDM (Mobile Device Management) enrolment · Separate work apps
- NEVER allow personal apps to sync to work accounts — Facebook, games, and social apps should not be on work devices

Data Confidentiality at Work

- NEVER share client records, financials, or strategy with unauthorised parties — including friends and family
- Data shared outside the organisation must use encrypted channels (VPN, secure email) with written authorisation
- NDAs (Non-Disclosure Agreements) are legally binding — violation can result in litigation and criminal liability under DPA 2021
- Default answer for 'should I share this data?' is NO — ask your supervisor if unsure

5 Essential Collaboration Tools Every CS Graduate Should Master

Microsoft Teams

Video conferencing, file sharing, persistent chat, integration with M365

Slack

Team messaging, channels, integrations with GitHub, Jira, and CI/CD tools

Zoom

Client meetings, webinars, screen sharing — still widely used in Zambia

Google Workspace

Docs, Sheets, Drive — real-time collaborative document editing

Jira / Trello

Project management, sprint tracking, issue management in software teams

Distracted Driving — A Digital Citizenship Issue

Distracted driving: Operating a vehicle while engaged with an electronic device — texting, calling, using social media, or checking navigation. Sending a text while driving increases crash risk by 23 times. (NHTSA)

What should I do with my device while driving?	What if I need to text or call someone urgently?	What else can I do for road safety?
• Turn it off or put it on silent	• Pull over safely and completely stop the vehicle first	• Secure children in appropriate car seats before starting
• Set up automated 'driving mode' responses that inform senders you are driving	• Ask a passenger to make the call or send the message for you	• Do not eat, drink, or apply makeup while driving
• Set up your GPS or maps app BEFORE you start driving	• Use voice-to-text ONLY if legally permitted and at a stop	• Avoid emotional conversations that distract mentally
• Place the phone face-down or in the glovebox	• Remember: no message is worth a life	• Never drive drowsy — pull over and rest instead

⚖️ **Zambia Legal Note:** Section 88 of the Zambia Road Traffic Act prohibits the use of a mobile phone while driving. Fines and licence suspension apply. Under the Cyber Security & Cyber Crimes Act 2021, causing death or injury through device-distracted driving may also carry criminal liability.

Zambia's ICT Legal Framework — Three Key Laws

Zambia has enacted three major pieces of legislation governing the digital environment. Every CS student must understand these laws — to protect themselves and comply with them professionally.

ICT Act, Cap. 469 (2009)

Establishes ZICTA

- Created the Zambia Information & Communications Technology Authority (ZICTA) as the principal regulator
- ZICTA licenses ISPs, mobile operators, and value-added service providers
- Manages the radio frequency spectrum; promotes rural ICT access
- Handles consumer complaints in the telecoms sector
- Sets technical standards for ICT systems
- Report cybercrime: cybercrime@zicta.zm | +260 211 221 220

Cyber Security & Cyber Crimes Act, No. 2 of 2021

Criminalises cyber offences

- Unauthorised access (hacking): up to 25 years imprisonment
- Cyberbullying: up to 5 years imprisonment or fine
- Electronic fraud: up to 15 years imprisonment
- Identity theft: up to 15 years imprisonment
- Harmful content distribution: up to 5 years imprisonment
- EVIDENCE: Electronic messages (WhatsApp, email) ARE admissible in Zambian courts

Data Protection Act, No. 3 of 2021

Governs personal data

- Applies to ANY organisation processing personal data of Zambian residents
- 8 Principles: Lawfulness · Purpose Limitation · Minimisation · Accuracy · Storage Limit · Integrity · Accountability · Transparency
- Organisations (including developers who built the systems) face fines and criminal liability for non-compliance
- As a CS developer, you are legally obligated to build compliant systems
- Data Subject Rights: Access · Rectification · Erasure · Object · Portability · Complaint

MODULE 1 · SECTION D

Digital Footprint, Privacy & Responsibility

Active vs. Passive Footprint · Data Protection · Green Computing · Assistive Technologies

Your Digital Footprint

Every action you take online leaves a trace. Collectively, these constitute your digital footprint — largely permanent, even when you 'delete' content.

ACTIVE Footprint *Data you DELIBERATELY create and publish online*	**PASSIVE Footprint** *Data collected about you WITHOUT explicit action on your part*
‣ Social media posts, comments, likes, shares	‣ Websites you visit (tracked via cookies even in 'private' mode)
‣ Profile photos and bios	‣ Your GPS location (phone logs it constantly via apps)
‣ Forum contributions and online reviews	‣ Search history (recorded by search engines and ISPs)
‣ Emails and messages you send	‣ App usage patterns and in-app behaviour
‣ Files you upload, publish, or submit	‣ Mobile money and banking transaction metadata

7 Actions You Can Take TODAY to Manage Your Digital Footprint

🔍 Google yourself monthly — search your full name + 'CBU' + your city. This is what recruiters see.
🔒 Audit privacy settings monthly — platforms change defaults; what was private may now be public.
🔑 Use strong, unique passwords (12+ chars, mix of types) — use Bitwarden (free, open-source) to manage them.
📱 Enable 2FA (Two-Factor Authentication) on ALL email, banking, social, and academic accounts.
🔎 Audit app permissions — revoke access to camera, microphone, location for apps that don't need it.
🔏 Look for HTTPS before entering personal data — the padlock in the browser address bar is your signal.
💡 Remember: 'free' platforms (Facebook, Google, TikTok) monetise YOUR data. You are not the customer — you are the product.

Zambia Data Protection Act 2021 — 8 Principles & Rights

The 8 Data Protection Principles

1	**Lawfulness:**	Data collected only with legitimate legal basis — usually explicit, informed consent
2	**Purpose Limitation:**	Data collected for one purpose cannot be used for a different, incompatible purpose
3	**Data Minimisation:**	Collect ONLY the minimum data necessary. Do not collect data 'just in case'
4	**Accuracy:**	Personal data must be kept accurate and up to date; inaccurate data must be corrected
5	**Storage Limitation:**	Data must not be retained longer than necessary. Delete when no longer needed
6	**Integrity & Confidentiality:**	Protected against unauthorised access, loss, or destruction through security measures
7	**Accountability:**	The organisation is responsible for compliance and must be able to demonstrate it
8	**Transparency:**	Data subjects must be clearly informed about collection, purpose, use, and their rights

Data Subject Rights

Right to be Informed:
Know what data is collected and why

Right of Access:
Request a copy of YOUR data from any org

Right to Rectification:
Have inaccurate data corrected

Right to Erasure:
'Right to be Forgotten' — deletion in certain cases

Right to Object:
Stop your data being used for marketing

Right to Portability:
Receive your data in a usable format

Right to Complain:
Lodge formal complaint with Data Protection Commissioner

⚠️ FOR CS DEVELOPERS: You are legally obligated to build systems that comply with the DPA 2021 by design. This means: consent mechanisms · minimum data collection · encryption at rest and in transit · user access and deletion tools · audit logs. Non-compliance: fines + criminal liability.

Green Computing & Electronic Waste

Green Computing: The practice of reducing electricity consumed and environmental waste generated when using computers, mobile devices, and related technologies. (Campbell, 2023) **E-Waste:** Discarded electronic equipment — one of the fastest growing waste streams globally.

Individual Green Computing Habits	Organisational Green IT Measures
✓ Purchase devices with the ENERGY STAR certification label	✓ Server consolidation: run multiple virtual servers on one physical server
✓ Shut down computers and devices overnight or when not in use	✓ Purchase high-efficiency equipment and power supplies (80 Plus certified)
✓ Use sleep mode instead of screensavers — screensavers waste power	✓ Use sleep modes and power management features across all devices
✓ Avoid replacing devices every time a new version is released	✓ Recycle or properly dispose of obsolete technology through certified e-waste handlers
✓ Use paperless communication — digital documents, e-signatures	✓ Use outside air (free cooling) where possible for data centre temperature management
✓ Print only when essential; use double-sided printing	✓ Allow employees to telecommute to reduce transport emissions
✓ Recycle toner/ink cartridges, batteries, and old devices through certified recyclers	✓ Migrate to cloud services — shared infrastructure is more energy-efficient than dedicated hardware
✓ Use video conferencing instead of travelling for meetings	✓ Procure refurbished equipment where appropriate

📍 Zambia: E-waste from old phones and computers often ends up in informal dumps near Lusaka and Copperbelt. ZEMA (Zambia Environmental Management Agency) regulates e-waste disposal. Choose certified recyclers.

Assistive Technologies & Digital Accessibility

Accessibility: The practice of removing barriers that prevent individuals with disabilities from interacting with data, websites, or apps. **Assistive Technology:** Any device, software, or equipment that helps people work around their challenges. (Campbell, 2023)

👁 Visual Impairments

- Screen magnification: increase text size and zoom level in OS settings
- High contrast modes: change colour schemes to improve readability for low vision
- Screen readers: JAWS, NVDA (Windows), VoiceOver (Apple) — read screen content aloud
- Alt text: descriptive text for images that screen readers read aloud

👂 Hearing Impairments

- Visual alerts: apps display visual cues (flashes, banners) instead of audio sounds
- Closed captions / subtitles on all video content (now legally required in many countries)
- Video relay services (VRS): sign language interpreters connect via video call
- Speech-to-text: transcribes spoken audio to text in real time

✋ Motor / Mobility Challenges

- On-screen keyboards: touch or eye-tracking instead of physical keyboard
- Switch access: single-switch scanning for users who cannot use keyboard/mouse
- Voice recognition: Dragon NaturallySpeaking, Windows Speech Recognition
- Eye-tracking technology: gaze control replaces mouse input

🧠 Intellectual / Learning Disabilities

- Speech-to-text input: users dictate instead of type
- Text-to-speech: audio reading aloud for users with reading difficulties (dyslexia)
- Graphic organisers: visual outlines structure information for clarity
- Simplified interfaces: clear, consistent layouts with reduced cognitive load

🏗 DESIGN OBLIGATION: As CS developers, you must design accessible systems from day one (Accessibility by Design). WCAG 2.1 (Web Content Accessibility Guidelines) is the international standard.

Tutorial Discussion Activity — Module 1

Work in pairs or groups of 3 — 12 minutes total:

1	A WhatsApp message in your family group claims that the Zambia Revenue Authority (ZRA) will freeze all bank accounts of people who have not updated their NRC details online via a link provided. Apply SIFT. What are the red flags? 💡 *Think about who would legitimately communicate this way. What organisation sends such urgent requests via WhatsApp? Is the link verifiable?*	**SIFT + Identity Safety**
2	Your classmate asks you to write 50% of their Assignment 2 because they are overwhelmed. They promise to put both your names on it. What do you do, and which of the six pillars of academic integrity does this situation involve? 💡 *Consider ALL six pillars — not just honesty. What does responsibility require you to do? What does courage look like here?*	**Academic Integrity**
3	Your new IT employer asks you to share a spreadsheet containing 500 customer records (names, NRCs, phone numbers) via your personal Gmail account for a marketing campaign. Identify: (a) the Data Protection Act principles this would violate, and (b) what you should do instead. 💡 *Which DPA principles apply? What is the correct secure data transfer procedure? Is this an order you can legally comply with?*	**DPA 2021 + Workplace ICT**
4	List five green computing practices that CBU's ICT department should implement in its computer laboratories. For each, estimate the benefit (reduced power, reduced waste, reduced cost, or all three). 💡 *Think about the labs you already use. What runs 24/7? What is printed unnecessarily? What happens to old lab computers?*	**Green Computing**

Key Takeaways — Module 1

1

Digital citizenship is more than digital literacy

Literacy is about skills; citizenship is about ethical, legal, and responsible use. You already have the skills — now you must develop the judgement.

2

SIFT before you share — every time

Stop → Investigate → Find better coverage → Trace. One false forward from you can harm hundreds. Apply this to WhatsApp, Facebook, and email.

3

Academic integrity is non-negotiable

Plagiarism, contract cheating, and AI misuse have real consequences: failure, suspension, expulsion. The Golden Rule: if you cannot explain it, do not submit it.

4

Your digital footprint is permanent

Nothing online is truly deleted. Build your professional digital brand now — LinkedIn, GitHub, professional email. Google yourself today.

5

Zambia's laws are real and enforced

Cyber Crimes Act 2021 and Data Protection Act 2021 are active. Electronic messages ARE admissible as evidence. Your online actions have legal consequences.

6

Accessibility and environment are your responsibility

Design accessible systems from day one. Practise green computing. As a CS professional, the systems you build affect millions of people — including the most vulnerable.

Can You Now...? — Learning Outcomes Self-Check

After this module, honest self-assessment: can you do each of the following without referring to notes?

- ✓ Define digital citizenship and explain the difference between digital literacy and digital citizenship
- ✓ Name and explain all 9 of Ribble's Elements of Digital Citizenship with Zambian examples
- ✓ Explain the digital divide and its specific causes and implications in Zambia
- ✓ Distinguish misinformation, disinformation, and malinformation and apply the SIFT framework
- ✓ Describe at least 5 techniques used to steal personal information, and 4 ways to protect yourself
- ✓ Define netiquette and list 5 rules for professional email and 5 rules for social media
- ✓ Name the 6 pillars of academic integrity (ICAI) and define plagiarism and its 7 forms
- ✓ Explain CBU's AI use policy framework — what is permitted, what requires disclosure, what is forbidden
- ✓ List the key provisions of the CBU Acceptable Use Policy and the consequences of violation
- ✓ Describe 3 ways technology transforms the professional world (e.g. mHealth, CAM, telemedicine)
- ✓ Explain active vs. passive digital footprints and list 7 strategies to manage yours
- ✓ Name the 8 principles of the Data Protection Act 2021 and the 7 rights of data subjects
- ✓ Explain what green computing is and list 5 individual and 5 organisational practices
- ✓ Define assistive technology and give examples for visual, hearing, motor, and intellectual challenges

Assessment & What's Next

This Week's Tasks

- ✓ Complete tutorial (Module 1 content)
- ✓ Google yourself and audit your social media privacy settings
- ✓ Create or update your LinkedIn profile
- ✓ Create a GitHub account (even if you have no projects yet)

⭐ Assignment — Digital Citizenship Case Study

Task: Select a REAL digital citizenship incident in Zambia (cybercrime, data breach, misinformation event, or academic integrity violation). Analyse using Module 1 frameworks (SIFT, DIKW, Ribble's elements, Zambian law). Recommend prevention strategies.

Module 2 (Next: Foundations of Computing)

Topics coming next:

- 1.1 Data, Information, Knowledge & Wisdom (DIKW)
- 1.2 History & Evolution of Computing (5 generations)
- 1.3 Classification of Computers (supercomputers to IoT)
- 1.4 Moore's Law and its implications
- 1.5 Cloud Computing — introduction

Why Module 2 Matters:

- Every module in this course builds on the foundation of Module 2
- Number systems (Module 3) require understanding what 'data' is
- Hardware (Module 4) requires understanding why generations matter
- All CS courses assume you understand computer classification
- Cloud and IoT connect to your future career in every sector

References & Further Reading

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